Supporting Information for "Do flood susceptibility models know more than the flood record? Evidence from radar-observed floods in Bangladesh"

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This file gives the pipeline settings with a map of the two hazard surfaces (Fig. S1), the construction of the composite risk and the sensitivity of its weights, which the main text summarises. Section, figure and table numbers without the prefix S refer to the main text. The code that produces every number here is at `https://github.com/rakibhhridoy/rimes-mapathon`, and the data at `https://doi.org/10.5281/zenodo.23267002`.

S1 Pipeline and model settings

Infrastructure, terrain and features

Assets are fetched from the Overpass API by bounding box, in tiles of 0.75°, and clipped to the region afterwards. The boxes cross Bangladesh's land border, so assets in India, or on land more than about 1 km outside the geoBoundaries border, are then dropped, keeping those on chars, coastal ground and offshore, and the risk grid and the grid-level validation are restricted the same way. Tiling avoids the timeouts that a query for a whole division produces. Three tag groups are requested, with lifeline assets covering hospitals, clinics, schools, colleges, shelters, bridges, embankments and dams, transport assets covering primary to trunk roads, railways and tagged bridges, and agricultural assets covering farmland, aquaculture, canals, ditches and marketplaces. Each feature is classified into one asset type from its tags and labelled with the district it falls in.

The SRTM tiles covering the bounding box are merged, reprojected to UTM zone 46N and clipped. From the projected elevation model the pipeline derives slope, the topographic wetness index (TWI), height above nearest drainage (HAND) and D8 flow accumulation. Drainage cells for HAND are those above the 90th percentile of flow accumulation, and HAND is the elevation difference between each cell and the nearest drainage cell found by a distance transform.

For each asset the pipeline samples the five terrain rasters at its representative point, samples WorldPop density, and computes the distance in metres to the nearest hospital, school, flood shelter, road and irrigation channel. Asset type enters as a categorical code, all features are standardised to zero mean and unit variance, and the label is the flood-prone status of the ground under the asset. Sampling reprojects the query coordinates into each raster's coordinate system before reading it, and the pipeline stops if any feature comes out constant or if the labels contain a single class.

Asset model, graph and calibration

The model that scores assets is a configuration setting, `asset_model.type`, and the default is class-weighted gradient boosting over the asset features, at 300 boosting iterations with a learning rate of 0.1. That default follows from the comparison in Section 5.6, which found ordinary tabular models matching or beating the graph network the system began with. The graph network remains available, as a two-layer GraphSAGE with 64 hidden channels and dropout of 0.3, trained with a class-weighted loss and early stopping on the calibration blocks, and so do a random forest and logistic regression. In the two riverine regions the scores shown on the map come from the same model given one more feature, the share of

earlier Sentinel-1 events in which the ground flooded, trained to predict the flooding of the latest event year from the years before it and applied with the share over every event (Section 5.4). Every validation figure in this paper comes from the model without that feature, because the feature is built from the same events the models are scored against.

Assets also form the nodes of a k-nearest-neighbour graph with $k = 6$, where any pair further apart than 5 km, measured in metres in the projected coordinate system, is left unconnected. Inverse-distance weights are stored with each edge, although the GraphSAGE mean aggregation does not use them, so every retained neighbour counts equally. Under the default model the edges carry nothing and the graph serves only to hold the features, labels and block assignment that every model shares.

Validation holds out whole spatial blocks of 10 km instead of random nodes, because neighbouring assets share terrain, so a random split leaves every validation node next to training nodes that carry almost the same features and label, and the resulting score measures memorisation rather than skill. Blocks are assigned at random with a fixed seed to three groups, with 80 % for training and the remainder split evenly between calibration and test. Early stopping and the isotonic calibration of the scores use the calibration blocks only, and every metric reported in this paper comes from the test blocks, which neither has seen. Isotonic regression returns the observed frequency of each level it fits, so a level holding a handful of calibration points that all flooded returns a probability of one. Each level is therefore shrunk by its Jeffreys estimate, $(k + 0.5)/(n + 1)$ for k of n points flooded, which leaves a level carrying many points almost unchanged and pulls a level carrying few back towards the middle. In Rangpur and Rajshahi the raw scores give a Brier score of 0.069 on the test blocks and the calibrated probabilities 0.029, against 0.032 for always predicting the base rate, so the calibration corrects the inflated raw scores but adds little over the base rate. Flood prevalence differs from one set of blocks to another, which limits how far a calibration carries. In Rangpur and Rajshahi 6.4 % of the assets in the calibration blocks sit on flood-prone ground against 3.3 % in the test blocks, and the calibrated deciles nonetheless lie close to the diagonal there, while on the coast, with 5.5 % against 1.7 %, the calibration overstates the flooding in the test blocks (Fig. 5). The calibrated value is best read as an approximate probability for the region as a whole, reliable in its ranking but not in its level for any one area.

Hazard surfaces

Two hazard surfaces are produced for every flood region, and the composite risk uses one of them according to the configuration setting `risk.hazard_source`. The first is ordinary Kriging of the model scores at the asset locations, with a spherical variogram fitted on up to 3,000 points and a residual Kriging pass that adds back the score residuals at the nodes. How much it can say between assets depends on the spatial structure of the scores, which the fitted variogram reports. The nugget is the share of the variance that is unexplained at zero distance, and the mean Kriging variance over the full sill is the share the interpolation leaves unexplained on average. On the current runs the nugget is 0.307 of the sill in Rangpur and Rajshahi and 0.742 in Sylhet, and the mean Kriging variance is 0.442 and 0.933 of the sill respectively, so in both regions the interpolated surface smooths substantially towards the regional average. In Rangpur and Rajshahi its standard deviation across the grid is 0.127 against 0.238 for the terrain surface described next, and it varies smoothly over tens of kilometres where the terrain surface follows individual channels and levees (Fig. S1).

The second surface is a class-weighted logistic regression from each grid cell's own terrain, using elevation, slope, TWI, HAND and flow accumulation and nothing else, because distances to hospitals or roads describe exposure and vulnerability rather than the likelihood of flooding. It is trained at the asset locations on the same labels and the same spatial blocks as the asset model, and evaluated at every grid cell, so it varies at the resolution of the elevation model rather than at the spacing of mapped infrastructure. In Rangpur and Rajshahi it reaches an AUC of 0.621 on held-out blocks, and it is the default. The two surfaces can also be scored directly against observed flooding over every grid cell, not only at assets, and there the terrain surface reaches an AUC of 0.657 against 0.666 for the Kriged one. Both are written to the grid so the difference between them can be inspected. Because the terrain model is trained on the observed extents, it is a flood-probability surface fitted to what the radar saw. Were it trained on the proxy

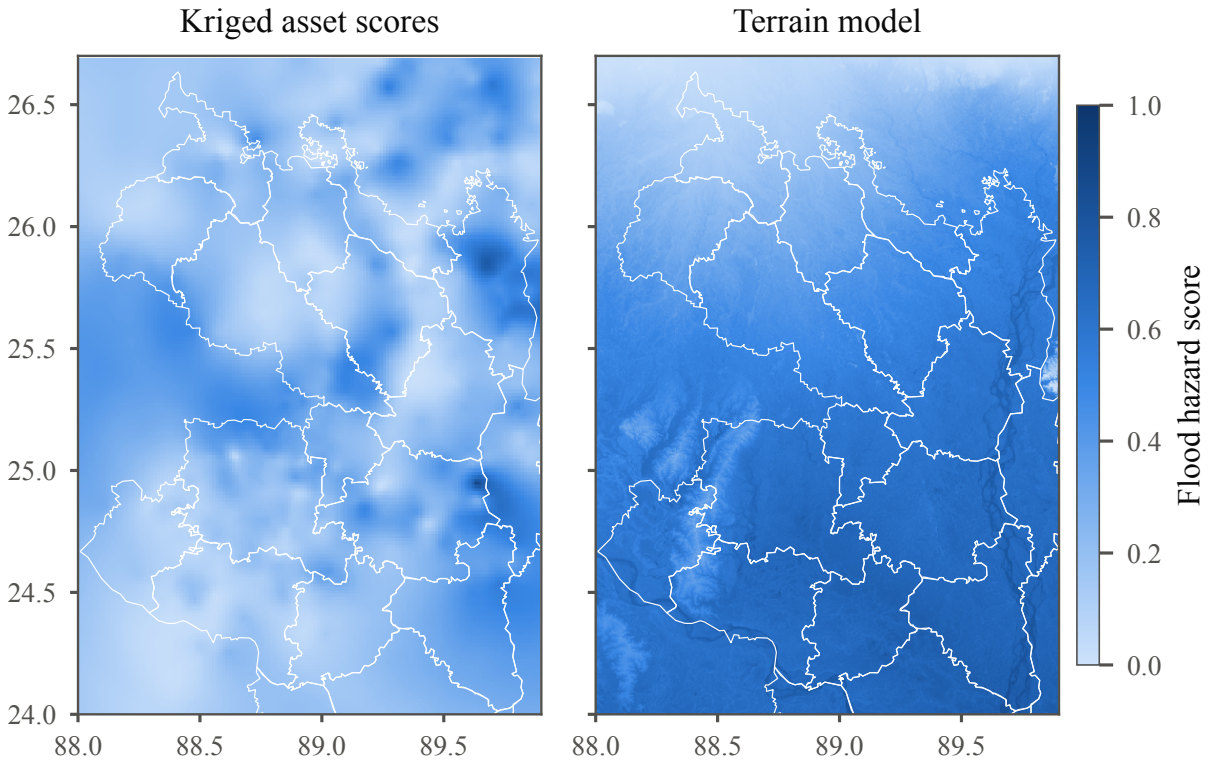


Figure S1: The two flood hazard surfaces for Rangpur and Rajshahi, with district boundaries in white. The Kriged surface interpolates the asset model's scores between mapped assets and has a standard deviation of 0.127 across the grid. The terrain model scores each 30 m cell from its own elevation, slope, TWI, HAND and flow accumulation, has a standard deviation of 0.238, and reaches a cell-level AUC against observed flooding of 0.657, against 0.666 for the Kriged surface.

labels instead, the circularity described in Section 4.3 would apply with full force, and it would be no more than ground the ensemble calls flood-prone, smoothed into a probability.

S2 Exposure, vulnerability, composite risk and aggregation

Exposure per cell is measured twice, because "where is infrastructure at risk" and "where would people be affected" are different questions with different answers. Infrastructure exposure is the type-weighted count of mapped assets in the cell, with hospitals and schools weighted at 0.20, bridges, roads and cropland at 0.15, shelters and embankments at 0.10 and other assets at 0.05, log-scaled to the unit interval. Population exposure is WorldPop density sampled at the cell centre, log-scaled the same way. Urban municipalities lead the first measure because that is where infrastructure is concentrated, while crowded floodplain settlements can lead the second.

Vulnerability combines population density with the distance from the cell centre to the nearest hospital, flood shelter and primary road, each normalised so that a greater distance means greater vulnerability. The configuration lists further components, a night-light proxy and an elderly-and-child ratio, that have no data source yet. Components without data are excluded rather than filled with a constant, and the remaining weights renormalised, and the weights actually applied are written to the output directory alongside the results. For the current runs the four applied components are population density (0.40), distance to hospital (0.27), distance to shelter (0.20) and distance to road (0.13).

Composite risk is the geometric mean of hazard, exposure and vulnerability, which keeps the multiplicative logic, in which no exposure means no risk, while staying on the unit scale of its inputs. Two composites are written, one for each exposure measure, and both are carried through to the administrative

summaries. Because the composite is bounded by its weakest factor it rarely approaches 1 even where risk is locally worst, so fixed cut-offs would place almost every cell in the lowest class. Cells are therefore also assigned relative classes 1 to 5 by quintile of the non-zero composite, the usual practice for susceptibility maps. Class 0 marks cells with no mapped exposure, which is an absence of data, not an absence of risk.

Grid results are summarised for every union, upazila and district from geoBoundaries, and each summary carries the mean and maximum of both composites, counts of exposed assets by type, and a flag stating whether the unit holds any mapped assets at all, and units without mapped assets have their risk left undefined. In Rangpur and Rajshahi 1,199 of the 1,339 unions hold mapped assets and carry a score. Unit names repeat across Bangladesh, so each unit also carries a label naming its parent, such as "Abdullahpur (Gangachara)". Hotspots are detected with the Getis–Ord G_i^* statistic on $k = 8$ nearest-neighbour weights, at 95 % confidence, over the cells that carry non-zero composite risk. Restricting the statistic to those cells keeps the weights matrix small enough to compute and avoids treating the empty majority of the grid as a sea of low values against which any occupied cell would look hot. Figure 3 maps the mean composite risk of every union in the three flood regions.

S3 Sensitivity of the composite to its weights

The composite risk rests on weights that were chosen rather than measured, so the ranking it produces is recomputed under other defensible choices and compared with the published one by Spearman correlation and by the share of the top decile of cells that stays in the top decile. Weighting every asset type equally leaves the ordering almost unchanged, at $\rho = 0.942$ in Rangpur and Rajshahi and $\rho = 0.908$ on the coast, and so does weighting the vulnerability components equally. Doubling the exposure exponent in the geometric mean gives $\rho = 0.946$ while retaining 0.829 of the top decile. Multiplying every chosen weight by a random factor between 0.5 and 1.5, over 20 draws, gives a mean ρ of 0.963 in Rangpur and Rajshahi, 0.956 in Sylhet and 0.978 on the coast, with 0.870 of the top decile retained.

Only one choice matters, reducing vulnerability to population density alone, which discards the distances to hospitals, shelters and roads and drops the correlation to 0.663 in Rangpur and Rajshahi and 0.425 on the coast, so the access distances rather than population drive the vulnerability term. The ordering of cells is therefore not an artefact of the particular weights, while the boundary of the top class is soft enough that a cell just inside it and a cell just outside should be treated alike.

S4 Validation metrics

Every model in the chain is scored on spatial blocks held out from training, and the same block assignment is shared by the asset model and the terrain hazard model within a region. Metrics are the area under the ROC curve, which measures ranking, average precision, which is more sensitive to the rare positive class, and the Brier score for calibration. Average precision is also reported as a lift over the base rate, because an average precision of 0.07 against a positive rate of 0.01 is a sevenfold improvement over chance while the same value against a rate of 0.10 is worse than chance.

The validation step then measures on the held-out blocks whether model scores rank assets that flooded above those that did not. Whether the hazard surface agrees with observed flooding across the whole grid, and not only where assets happen to be, is measured by AUC and Spearman correlation over all cells, and, because that includes the ground the surface was fitted on, also by AUC over the cells of the test blocks alone, with the surface refitted for each of the repeated block assignments. And how closely the proxy labels match observed flooding is measured with the categorical scores used in flood mapping, the probability of detection, false alarm ratio and critical success index, which quantifies what the proxy-trained models were learning.

Every metric that compares two models or two label sources is repeated over several block assignments drawn with different seeds, because a single assignment gives one number with no sense of its spread, and differences are tested by the corrected resampled t-test of Nadeau and Bengio (2003), which allows for the assignments sharing the same assets. Each such comparison uses 20 assignments.

Check	Rangpur & Rajshahi	Sylhet	South-west coast	Jamuna east
As published	0.801 / 0.896 ($p = 0.009$)	0.797 / 0.851 ($p = 0.036$)	0.807 / 0.668 ($p < 0.001$)	0.810 / 0.940 ($p = 0.090$)
Without always-flooded ground	0.752 / 0.815 ($p = 0.130$)	0.774 / 0.815 ($p = 0.185$)	0.806 / 0.667 ($p < 0.001$)	0.715 / 0.852 ($p = 0.187$)
Outside recorded water	0.762 / 0.833 ($p = 0.146$)	0.799 / 0.794 ($p = 0.912$)	0.827 / 0.661 ($p < 0.001$)	0.704 / 0.917 ($p = 0.096$)
Sentinel-1A masks only	0.795 / 0.891 ($p = 0.005$)	0.797 / 0.851 ($p = 0.036$)	0.802 / 0.649 ($p < 0.001$)	–
Target: any earlier flood	0.763 / 0.896 ($p < 0.001$)	0.834 / 0.851 ($p = 0.441$)	0.807 / 0.668 ($p < 0.001$)	0.835 / 0.940 ($p = 0.047$)
Target: flood fraction	0.765 / 0.896 ($p = 0.003$)	0.835 / 0.851 ($p = 0.401$)	0.720 / 0.668 ($p = 0.258$)	0.810 / 0.940 ($p = 0.114$)
20 km blocks	0.754 / 0.884 ($p < 0.001$)	0.795 / 0.850 ($p = 0.028$)	0.738 / 0.660 ($p = 0.182$)	0.693 / 0.868 ($p = 0.212$)
40 km blocks	0.750 / 0.876 ($p = 0.004$)	0.774 / 0.847 ($p = 0.017$)	0.729 / 0.676 ($p = 0.423$)	0.708 / 0.957 ($p = 0.023$)

Table S1: The model trained on floods up to 2022 against the record of earlier flooding, each scored by AUC against the 2024 floods on held-out blocks, as model / record, with the corrected resampled t-test of their paired difference over the block assignments. The validation region east of the Jamuna was not re-mapped from Sentinel-1A, and its few flooded assets leave only some assignments with both classes in the test blocks.

S5 Robustness of the comparison with the flood record

The temporal test of the main text trains the asset model on the floods up to 2022 and scores it, and the record of earlier flooding, against the ground the 2024 floods reached, on held-out 10 km blocks over 20 assignments. Table S1 repeats it under seven changes, each aimed at one way the comparison could favour the record or the model. The validation region east of the Jamuna is checked the same way. Its national region shares 1,263 mapped assets with the Rangpur and Rajshahi box, which crosses the Jamuna, and these are left out so that no asset is scored in both, leaving 4,152. Its October 2024 flood was a flash flood from the Meghalaya hills, and the first Sentinel-1 pass came five days after it began, so the fastest flooding in Sherpur may have drained unseen and the mask counts few flooded assets.

The first two checks leave the models alone and narrow the test set, because the record and the 2024 floods come from the same change detection, so ground that radar mistakes for water every monsoon, such as wet paddy, would count as flooded in both. Surfaces that stay dark in the dry season, such as sand bars and permanent wetland, are already rejected, since a pixel must darken against its dry-season baseline. The comparison is therefore repeated without the ground flagged in every earlier event, and without the ground the JRC record has ever seen under water, which removes seasonal wetland. Both removals also take away genuinely flood-prone ground, so they can only work against the record. In Sylhet the second leaves 72.0 % of assets, holding 29.9 % of the ground the 2024 floods reached.

The third check re-maps every event from Sentinel-1A alone, since Sentinel-1B failed in December 2021, so events before then were seen by two satellites and the 2024 events by one. Each event mask takes the lowest backscatter of its window, so a window with more scenes can map more flooding. With one satellite throughout, the earlier events shrank by up to 15.7 % in Rangpur and Rajshahi and 47.9 % on the coast, while in Sylhet Sentinel-1B had added no scenes and the masks are unchanged. The 2024 floods in Rangpur and Rajshahi covered 4.9–8.3 % of the region against 19.9–24.8 % for each earlier flood, and this difference is not a matter of scene counts, since the October 2024 and August 2017 windows hold the same number of Sentinel-1A scenes.

The fourth and fifth checks change what the model learns, since the published model learns a yes-or-no label, flooded in at least the region's minimum number of earlier events, while the record keeps the fraction of earlier events in which the ground flooded. Because the 2024 floods are counted as flooded in any of them, the model is retrained on flooding in any earlier event, and, as a regression, on the fraction itself. On the coast the first of these is the published target, since one event already suffices there.

The last two widen the blocks, since the 10 km blocks are narrower than the fitted variogram ranges of the asset scores (main text, Section 5.5). With 40 km blocks in Sylhet, 15 of the assignments leave both classes in the test blocks.

	Rangpur & Rajshahi	Sylhet	South-west coast	Jamuna east
AUC against the 2024 floods, record / model, by number of earlier events				
One event	0.812 / 0.768	0.726 / 0.794	0.579 / 0.699	0.845 / 0.811
Two events	0.873 / 0.788	0.851 / 0.797	0.634 / 0.794	0.907 / 0.786
Three events	0.896 / 0.801	–	0.668 / 0.807	0.940 / 0.810
Skill over the record				
Model	−0.905	−0.364	0.416	−2.152
Model given the record	0.276	−0.016	−0.063	0.327
Share of assets flooded in 2024 held by each list (%), record / model / model given the record				
All assets	93.2 / 68.5 / 93.2	77.9 / 50.6 / 62.0	37.1 / 9.5 / 23.8	88.2 / 21.6 / 84.3
Facilities	36.4 / 18.2 / 45.5	40.6 / 3.1 / 6.2	80.0 / 0.0 / 73.3	50.0 / 25.0 / 25.0
Transport links	88.3 / 34.0 / 88.3	71.6 / 56.8 / 32.8	38.9 / 0.0 / 33.3	87.5 / 12.5 / 81.2
Out-of-fold AUC, record / model / model given the record, and 95 % block-bootstrap interval of model minus record				
AUC	0.942 / 0.857 / 0.950	0.855 / 0.800 / 0.865	0.679 / 0.776 / 0.722	0.925 / 0.825 / 0.938
Interval	−0.127 to −0.064	−0.090 to −0.019	0.039 to 0.156	−0.191 to 0.003
Assets flooded in 2024	644	2,367	871	51

Table S2: Record length, skill over the record and planning lists. Record-length AUCs are means over every subset of that many earlier events and over the block assignments; Sylhet has two events before 2022. Skill is the share of the record's remaining ranking error that a model removes. The lists hold as many assets as an earlier flood reached in each region, with every asset scored out of fold.

S6 Record length, skill over the record and planning lists

Three further analyses ask what the record offers a flood manager, using the same events and the same temporal split as the main text. The first rebuilds both the record and the model from every subset of k earlier events, for k from one to all of them, and scores each on the held-out 10 km blocks against the 2024 floods over 20 assignments, with the model's label following the region's rule capped at k. A record of one event can only say flooded or not, so it ranks coarsely, while the model still ranks every asset. Rebuilt from a single earlier flood, the record already led the model in Rangpur and Rajshahi, whereas in Sylhet it needed both earlier floods, and on the coast it trailed at every length (Table S2).

The second expresses each model's AUC as skill over the record, $(\mathrm{AUC}_{\mathrm{model}} - \mathrm{AUC}_{\mathrm{record}})/(1 - \mathrm{AUC}_{\mathrm{record}})$, the share of the record's remaining ranking error that the model removes, in the way weather forecasts are scored against persistence. It is computed from the mean AUCs over the block assignments, because a ratio taken per assignment swings without bound where the record is nearly perfect. A score of zero means the model only matches the record, and a negative score that it falls behind.

The third turns the rankings into the list a planner would act on. Every asset is scored once out of fold, by models trained on the other nine of ten folds of 10 km blocks, so the whole region carries held-out scores. The record's own list is every asset an earlier flood reached, 15.5 % of assets in Rangpur and Rajshahi, 20.8 % in Sylhet and 2.5 % on the coast, and the model and the model given the record each list the same number of their highest-scoring assets. Each list is scored by the share of the assets the 2024 floods reached that it holds. Facilities are hospitals, schools and flood shelters, and transport links are roads, bridges and railways. Facilities flooded in 2024 number only 11, 32 and 15 in the three regions, so their shares are indicative only. Ranking the out-of-fold scores within each fold rather than across the region did not change which list held more of the flooded assets in any region. On the coast the record's lead holds for short lists only, since the top 5 % of assets it ranks held 38.6 % of the flooded assets against 16.0 % for the model, whereas the top 30 % held 54.4 % against 66.8 %, consistent with the model's higher AUC over the region as a whole. The uncertainty of the out-of-fold comparison is estimated by resampling whole 10 km blocks 1,000 times, so that neighbouring assets stay together. The interval for the model's AUC minus the record's excludes zero in every study region, below it in the riverine regions and above it on the coast, and east of the Jamuna it reaches 0.003, so the lead there is probable rather than established.

References

Nadeau, C. and Bengio, Y. (2003). Inference for the generalization error. *Machine Learning*, 52(3):239–281.